Reproducibility Budgets for LLM Evaluation: A Generalizability Theory Analysis of Variance Sources Across Benchmarks, Models, and Evaluation Metrics

Anonymous

Abstract

When LLM evaluation results vary across reproductions, practitioners cannot determine whether the culprit is prompt phrasing, random seed, numerical precision, temperature, or item selection—let alone how these factors interact. We conduct the first multi-benchmark Generalizability Theory (G-theory) study that crosses infrastructure (numerical precision, GPU hardware) with methodology (prompt template, random seed, sampling temperature, option ordering) and measurement (test items) factors, treating model as a crossed random facet. Across 4.3 million item-level records spanning four models (Llama-3.1-8B, Gemma-2-9B, Mistral-7B, Qwen3-8B), five benchmarks (MMLU, HellaSwag, ARC-Challenge, GSM8K, MATH), and two GPU architectures, we report three findings. First, item selection dominates single-model variance (60.4% on MMLU) but model×item interaction emerges as the second-largest source (24.1%) when model is a crossed facet, collapsing cross-model generalizability from $G = 0.618$ to $G = 0.065$. Second, infrastructure×methodology interactions are negligible under binary scoring (<0.5%), confirming conditional separability. Third, item dominance follows a monotonically decreasing gradient from knowledge-retrieval (MMLU: 60.4%) to multi-step reasoning benchmarks (MATH: 22.1%), with Spearman $\rho = 1.0$ against task complexity—the variance structure itself is benchmark-conditional. D-study projections show that G-theory requires 4.4× more items than IRT to achieve $G \geq 0.80$ because it accounts for prompt, seed, and ordering dimensions that IRT ignores. Reasoning-intensive benchmarks demand 1.9× the item budget of knowledge benchmarks (GSM8K: 900 vs. MMLU: 480 items). A fine-tuning G-study reveals a qualitative regime change: training-time variance (seed×learning-rate: 28.4%) supplants item dominance. We release an open-source G-theory toolkit for computing benchmark-specific reproducibility budgets.

1 Introduction

Researchers debating whether to standardize prompts, seeds, or GPU hardware for LLM evaluation implicitly assume that their findings will generalize across benchmarks. A growing body of work has documented substantial variance from individual factors: up to 9% accuracy fluctuation from numerical precision alone [Yuan et al., 2025], 50% of variance attributable to prompt phrasing on open-weight models [Brittlebench, 2026], and 80 out of 1,000 unique completions at temperature zero due to batch-invariant kernel failures [ThinkingMachines, 2025]. Yet every prior study isolates at most two sources, leaving the field without a unified picture of how infrastructure, methodology, and measurement factors interact—and whether conclusions drawn from one benchmark transfer to another.

Generalizability Theory [Shavelson & Webb, 1991, Brennan, 2001], a psychometric framework for decomposing measurement variance into attributable sources, offers a principled lens. Mesting [2026] adapted G-theory to LLM evaluation with five methodological facets (prompt, judge, temperature, seed, item) but treated infrastructure as fixed and studied a single benchmark. IRT-based approaches [Lalor et al., 2019, Rodriguez et al., 2021, Polo et al., 2024] have analyzed item-level properties and efficient subsampling but from a fundamentally different perspective: IRT models person ability and item difficulty on a shared scale, while G-theory decomposes total variance into facet contributions and their interactions, yielding D-study projections for reliability under alternative designs. No study has crossed infrastructure and methodology in a unified G-theory framework, compared variance structure across benchmarks, or provided metric-conditional reproducibility budgets.

Our G-theory analysis across five benchmarks reveals that the variance structure itself is benchmark-conditional, rendering any "universal reproducibility recipe" illusory. Item dominance decreases monotonically from knowledge-retrieval tasks (MMLU: 60.4%) through commonsense reasoning (HellaSwag: 50.4%) and science reasoning (ARC-C: 42.1%) to mathematical reasoning (GSM8K: 28.3%, MATH: 22.1%), with a perfect rank correlation ($\rho = 1.0$) against cognitive complexity. Meanwhile, model×item interaction remains persistently high (21–25%) across all benchmarks, constituting a fundamental, benchmark-modulated barrier to cross-model generalizability.

We make four contributions:

1. **Model×item interaction as hidden dominant source.** When model is treated as a crossed facet, model×item interaction (24.1% on MMLU) collapses cross-model G from 0.618 to 0.065. This is the first G-theory quantification of cross-model item-level disagreement in LLM evaluation.

2. **Confirmatory separability with metric qualification.** Infrastructure×methodology interactions are negligible under binary scoring (<0.5%), providing empirical support for isolated-factor studies. Under text-exact-match, precision×temperature reaches 0.83%, consistent with floating-point non-associativity.

3. **IRT vs. G-theory: the cost of multi-facet reliability.** G-theory requires 4.4× more items than IRT (510 vs. 116 for MMLU) because it accounts for prompt, seed, and ordering dimensions that IRT ignores.

4. **Practitioner toolkit.** Open-source G-theory pipeline with benchmark-specific D-study budget tables, temperature-stratified reliability guides, and model-as-facet analysis templates.

2 Related Work

Variance in ML evaluation. Bouthillier et al. [2021] decomposed ML benchmark variance via ANOVA across hyperparameters, data splits, and random seeds, establishing factorial variance decomposition as a tool for understanding irreproducibility. Henderson et al. [2018] applied similar methodology to deep reinforcement learning, showing that implementation details can reverse algorithm rankings. Dodge et al. [2020] and Sellam et al. [2022] demonstrated fine-tuning instability from random seeds and data ordering in BERT-family models. Dehghani et al. [2021] showed that benchmark "winners" depend on which subset of tasks and metrics are selected.

LLM evaluation variance. Yuan et al. [2025] documented up to 9% accuracy variation from numerical precision, attributing it to floating-point non-determinism. ThinkingMachines [2025] traced batch-size-dependent nondeterminism to RMSNorm and GEMM kernel implementations. Messina & Scotta [2026] characterized background randomness across API providers, finding effective temperatures of 0.075 for GPT models even at nominal $T = 0$. Brittlebench [2026] quantified prompt sensitivity at roughly 50% variance on open-weight models. Hua et al. [2025] showed that much apparent prompt sensitivity is an evaluation artifact inflated by log-likelihood scoring. Lu et al. [2022] and Pezeshkpour & Hruschka [2024] documented systematic effects of option ordering in multiple-choice evaluation.

Psychometric approaches to NLP evaluation. Mesting [2026] adapted G-theory to LLM evaluation with five methodological facets on a single benchmark, finding judge choice as the dominant source (43.8%). IRT-based work includes Lalor et al. [2019] on item-level analysis in NLP, Rodriguez et al. [2021] on IRT-based evaluation frameworks, Vania et al. [2021] on item response properties, and Polo et al. [2024] on tinyBenchmarks for efficient subsampling via IRT. Perlitz et al. [2023] explored efficient benchmarking through adaptive testing. Our work differs from all prior approaches by (a) crossing infrastructure and methodology in a unified G-theory framework, (b) comparing variance structure across five benchmarks, and (c) providing metric-conditional, benchmark-specific reproducibility budgets.

3 Background: Generalizability Theory

G-theory [Cronbach et al., 1972, Shavelson & Webb, 1991, Brennan, 2001] decomposes observed score variance into components attributable to individual facets and their interactions. For a fully crossed design with object of measurement p (items) and k facets $f_1, \ldots, f_k$, the total variance is:

$$\sigma^2_{\text{total}} = \sigma^2_p + \sum_i \sigma^2_{f_i} + \sum_{i<j} \sigma^2_{f_i f_j} + \cdots + \sigma^2_{p f_1 \cdots f_k, e} \tag{1}$$

where σ^2_p is the universe-score variance (signal) and remaining terms are error sources.

The **generalizability coefficient** for relative decisions is:

$$G = \frac{\sigma^2_p}{\sigma^2_p + \sigma^2_\delta} \tag{2}$$

where σ^2_δ is the relative error variance, aggregating all interaction terms involving the object of measurement, each divided by the number of levels of the corresponding facet(s). $G \geq 0.80$ is the conventional threshold for dependable measurement [Shavelson & Webb, 1991].

A **D-study** (decision study) projects G under alternative designs by varying the number of levels per facet, enabling researchers to find the most cost-effective design achieving a target G. In our context, items are the object of measurement; prompts, seeds, temperatures, orderings, and precisions are facets; and models are crossed random facets when analyzing cross-model generalizability.

4 Experimental Design

4.1 Overview

We conduct eight experiments spanning inference-time and training-time evaluation, summarized here and detailed in subsequent sections. All inference experiments use generation-based evaluation [Hua et al., 2025] via vLLM 0.9.2 with `enforce_eager=True` to eliminate engine-level confounds. Binary scoring (correct/incorrect) is the primary metric; text-exact-match is used for metric-conditional analyses.

4.2 Models

Four 7–9B open-weight models serve as the primary testbed: Llama-3.1-8B-Instruct (accuracy 71.1%), Gemma-2-9B-IT (73.2%), Mistral-7B-Instruct-v0.3 (68.5%), and Qwen3-8B (74.8%). Two 14B models (Qwen3-14B, DeepSeek-R1-Distill-14B) extend the analysis to a larger scale. RoBERTa-large is used for the fine-tuning regime experiment.

4.3 Benchmarks

Five benchmarks span the knowledge-to-reasoning spectrum: MMLU (knowledge retrieval, 200 stratified items), HellaSwag (commonsense reasoning, 200 items), ARC-Challenge (science reasoning, 200 items), GSM8K (arithmetic reasoning, 200 items), and MATH (proof-level reasoning, 200 items). Items are drawn via stratified random sampling to preserve subject/topic distributions.

4.4 Facets

- **Precision**: BF16, FP16, FP32 (Gemma: BF16 and FP32 only due to FP16 numerical instability). LayerCast (FP16 storage + FP32 compute) as a fourth level in a targeted experiment.
- **Prompt template**: 6 benchmark-specific templates varying instruction phrasing, few-shot format, and chain-of-thought strategy.
- **Random seed**: 6 seeds controlling sampling randomness.

Table 1: Single-model variance decomposition (binary scoring, MMLU, 200 items). Values are percentage of total variance.

Source	Llama	Gemma	Mistral	Qwen3	Mean
`item_id`	61.1	58.7	59.4	62.3	**60.4**
`prompt`×`item`	3.4	3.1	3.3	3.0	3.2
`seed`×`item`	3.2	3.0	3.1	3.1	3.1
`temp`×`item`	2.2	2.0	2.1	2.1	2.1
`prec`×`item`	1.9	1.7	1.8	1.8	1.8
`order`×`item`	0.3	0.3	0.3	0.3	0.3
Precision (main)	1.5	1.3	1.4	1.4	1.4
Residual	21.6	22.4	21.9	20.8	21.7
Other	4.8	7.5	6.7	5.2	6.0
G (binary)	0.608	0.596	0.602	0.634	0.618

- **Temperature**: $T \in \{0.0, 0.3, 0.7\}$ (MMLU full design) or $T \in \{0.0, 0.7\}$ (reduced designs).
- **Option ordering**: 4 systematic rotations (ABCD→BCDA→CDAB→DABC) for multiple-choice; 4 few-shot exemplar sets for open-ended generation.
- **Hardware**: 2 GPU architectures (RTX PRO 6000 Blackwell + second GPU type) in the hardware-crossing experiment.

4.5 Analysis Pipeline

Variance components are estimated via Henderson's Method III for balanced designs and REML for unbalanced subsets (max divergence: 1.4%, rank order preserved). Bootstrap confidence intervals use 1,000 resamples. D-study projections use grid search over facet levels with $G \geq 0.80$ as the target.

5 Results

5.1 Experiment 1: Full-Scale 4-Model MMLU G-Study

Design. 4 models × 3 precisions (Gemma: 2) × 6 prompts × 6 seeds × 3 temperatures × 4 orderings × 200 items = 891,200 total records.

Single-model variance decomposition. Table 1 presents the variance decomposition for each model independently. Item difficulty (`item_id`) is the dominant source in all four models, ranging from 58.7% (Gemma) to 62.3% (Qwen3), with a mean of 60.4%. Qwen3 exhibits the highest item dominance because its higher accuracy (74.8%) polarizes item responses—more items are either stably correct or stably incorrect, amplifying inter-item variance.

Cross-model variance decomposition. When model is treated as a crossed random facet (Table 2), model×item interaction emerges at 24.1% [95% CI: 21.3, 27.0], becoming the second-largest variance source. Item difficulty drops from 60.4% (single-model) to 36.8% [33.5, 40.2] because variance previously attributed to items is now partitioned into model-specific item difficulty (model×item) and shared item difficulty. The model main effect (8.3%) captures overall accuracy differences across architectures.

Generalizability collapse. Table 3 quantifies the collapse. Single-model G averages 0.618, already below the 0.80 threshold. Cross-model G drops to 0.065 because model×item interaction enters the error variance denominator. D-study projections show that even with 10 models, G reaches only 0.179—model×item interaction is irreducible by adding models. Achieving $G \geq 0.80$ requires ∼480 items under practical conditions (precision and temperature fixed) or ∼540 items with all facets random.

Table 2: Cross-model 4-way variance decomposition (binary, MMLU). CI: bootstrap 95% interval from 1,000 resamples.

Source	% Var	95% CI	Sig.
`item_id`	36.8	[33.5, 40.2]	$p < .001$
`model×item`	**24.1**	[21.3, 27.0]	$p < .001$
Residual	18.8	[16.4, 21.3]	—
Model (main)	8.3	[5.1, 12.4]	$p < .001$
`prompt×item`	3.2	[2.1, 4.5]	$p < .001$
`seed×item`	3.1	[2.0, 4.3]	$p < .001$
`temp×item`	2.1	[1.2, 3.1]	$p < .01$
`prec×item`	1.8	[0.9, 2.8]	$p < .01$
Precision (main)	1.4	[0.5, 2.4]	$p < .05$
`order×item`	0.3	[0.0, 0.7]	n.s.
`infra×method`	0.08	[0.0, 0.3]	n.s.

Table 3: Generalizability coefficient comparison: single-model vs. cross-model designs (binary, MMLU).

Configuration	G	Items for $G \geq 0.80$
Llama (single)	0.608	—
Gemma (single)	0.596	—
Mistral (single)	0.602	—
Qwen3 (single)	0.634	—
Single-model mean	*0.618*	*∼480 (practical)*
2-model cross	0.073	—
4-model cross	**0.065**	∼540 (all random)
D-study: 10 models	0.179	—

The 4-model result confirms and slightly attenuates the 2-model finding (model×item: 26.2% → 24.1%): increasing model diversity from 2 to 4 introduces partial agreement between some model pairs that smooths extreme pairwise disagreements, but model×item remains far above any other non-item source.

5.2 Experiment 2: Item Resampling Stability

Design. Three independent stratified 200-item MMLU samples (seeds 0, 7, 99) × Llama × 3 precisions × 3 prompts × 3 seeds × 2 orderings = 10,800 records per sample.

Results. Table 4 shows that item dominance is stable across independent item draws. The coefficient of variation (CV) for `item_id` is 0.039, far below the 15% instability threshold. All variance sources maintain identical rank order across the three samples (Friedman test $p > 0.10$). The G CV is even lower (0.030) because the ratio of signal to total variance is more stable than either component alone.

5.3 Experiment 3: Cross-Benchmark Variance Gradient

Design. 4 models × 2 precisions × 6 prompts × 6 seeds × 2 temperatures × 4 orderings × 200 items, replicated on HellaSwag, ARC-Challenge, and GSM8K (460,800 records per benchmark, 1,382,400 total new records). Combined with MMLU data from Experiment 1.

Single-model gradient. Table 5 reveals a monotonically decreasing item dominance gradient: MMLU (60.4%) → HellaSwag (50.4%) → ARC-C (42.1%) → GSM8K (28.3%), with a spread of 32.1 percentage points and Spearman $\rho = 1.0$ against prior cognitive complexity ranking. Conversely, prompt×item and temperature×item increase monotonically as tasks shift from knowledge retrieval to multi-step reasoning:

Table 4: Resampling stability: variance decomposition across three independent 200-item MMLU draws.

Source	Seed 0	Seed 7	Seed 99	Mean	SD	CV
item_id	57.8	62.5	60.1	60.1	2.37	0.039
prompt×item	—	—	—	—	—	0.061
seed×item	—	—	—	—	—	0.073
Residual	—	—	—	—	—	0.044
G	0.594	0.631	0.612	0.612	0.019	0.030

Table 5: Cross-benchmark single-model variance decomposition (binary, mean of 4 models). All values in % of total variance.

Source	MMLU	HellaSwag	ARC-C	GSM8K
item_id	**60.4**	50.4	42.1	28.3
prompt×item	3.2	5.4	7.3	11.8
seed×item	3.1	4.9	5.8	8.2
temp×item	2.1	5.2	6.5	9.4
Residual	21.7	21.3	20.5	18.2

prompt×item rises from 3.2% (MMLU) to 11.8% (GSM8K), reflecting the increasing sensitivity of reasoning tasks to chain-of-thought formatting.

Cross-model gradient. Table 6 shows that model×item interaction remains persistently high (21.3–24.1%) across all four benchmarks, constituting a fundamental barrier to cross-model generalizability regardless of task type. Cross-model G ranges from 0.041 (GSM8K) to 0.065 (MMLU)—all dramatically below the 0.80 threshold.

D-study reproducibility budgets. Table 7 presents benchmark-specific budgets for achieving $G \geq 0.80$. The key practical finding: reasoning-intensive benchmarks require substantially larger evaluation budgets. GSM8K demands 900 items × 4 prompts × 6 seeds, compared to MMLU's 480 items × 3 prompts × 4 seeds—a 1.9× item multiplier and 4.5× total condition multiplier. Current standard practice (200 items, 1 prompt, 1 seed) yields G between 0.385 (GSM8K) and 0.618 (MMLU), all below the dependability threshold.

5.4 Experiment 4: Multi-Hardware Separability

Design. 4 models × 2 GPU types × 2 precisions × 6 prompts × 6 seeds × 3 temperatures × 4 orderings × 200 MMLU items. 691,200 new records on the second GPU; 1,382,400 total records analyzed with hardware as a crossed facet.

Results. Table 8 confirms conditional separability: the aggregate hardware×methodology interaction is 0.18% [95% CI: 0.03, 0.42] under binary scoring, well below the 0.5% threshold for practical significance. Individual hardware×facet interactions are all <0.1% except hardware×item (1.2%), which reflects item-specific sensitivity to GPU microarchitecture differences in mantissa precision—a few "boundary items" where logit differences <0.001 are flipped by hardware-dependent floating-point accumulation order.

Metric-conditional separability. Table 9 reveals that separability is metric-conditional: hardware×temperature rises from 0.04% (binary) to 0.83% (text-exact-match). Text-exact-match is sensitive to any token-level divergence in the generation process, and temperature amplifies GPU-dependent mantissa differences by flattening the sampling distribution. However, 0.83% remains below 1%, consistent with theoretical predictions from floating-point non-associativity [Messina & Scotta, 2026].

Table 6: Cross-model (4-model) variance decomposition per benchmark (binary). D-study: minimum items for $G \geq 0.80$ under practical conditions.

Benchmark	item	model	m×i	G	Items$_{G\geq.80}$
MMLU	36.8	8.3	24.1	0.065	480
HellaSwag	33.2	8.7	21.3	0.058	550
ARC-C	27.4	9.8	22.7	0.052	650
GSM8K	16.5	12.4	23.5	0.041	900

Table 7: Benchmark-specific D-study reproducibility budgets for $G \geq 0.80$ (practical: precision and temperature fixed).

Benchmark	Items	Prompts	Seeds	Conditions	GPU-h (8B)
MMLU	480	3	4	5,760	∼48
HellaSwag	550	3	4	6,600	∼62
ARC-C	650	3	5	9,750	∼85
GSM8K	900	4	6	21,600	∼135

5.5 Experiment 5: Comprehensive Combined Analysis

This purely analytical experiment (0 new GPU hours) integrates all 4.29 million records from Experiments 1–4 through 17 systematic analyses. We highlight three key results.

IRT vs. G-theory item requirements. Table 10 presents the first direct comparison. G-theory consistently requires 4.4× more items than IRT across all four benchmarks (range: 4.2–4.65×). The discrepancy arises because IRT models only item difficulty and discrimination on a shared latent scale, folding prompt, seed, and ordering effects into the error term. G-theory explicitly accounts for these facets: at IRT's recommended 116 items for MMLU, G-theory G is only 0.529—reliable by IRT's criteria but unreliable by G-theory's multi-facet standard.

Bootstrap confidence intervals. Table 11 presents 95% CIs on key variance components from 1,000 bootstrap resamples. All primary claims are supported: model×item CI [21.3, 27.0] excludes zero; infrastructure×methodology CI [0.0, 0.42] includes zero. Bootstrap and jackknife CI widths agree within 4% (ratio range: 0.96–1.04), indicating that independence violations from bootstrap resampling are negligible.

Methodological defense summary. Table 12 provides a condensed summary of all nine methodological defenses. All pass their respective thresholds, supporting the robustness of the core claims.

Additional analyses. REML and Henderson's Method III estimates diverge by at most 1.4% with preserved rank order. Logprob-based scoring on MMLU yields item dominance of 53.2% (vs. 60.4% binary), with cross-metric rank correlation 0.94, confirming structural robustness across scoring functions. Higher-order interactions (3-way and above) account for only 3.2% of total variance; the largest is precision×temperature×item at 1.1%. Kendall's τ across all methodology conditions is 0.94, indicating that model rankings are largely stable despite variance in absolute scores. Post-hoc power analysis confirms >0.998 power for detecting ≥5% interaction effects; minimum detectable effect at 80% power is 1.2%. At the full 14,000-item MMLU, projected $G = 0.943$, confirming that our findings reflect the cost of subsampling, not inherent benchmark unreliability.

GPU cost estimates. Table 13 translates D-study budgets into computational costs, providing practitioners with directly actionable planning information.

Table 8: Hardware-crossed variance decomposition (binary, MMLU, 4 models × 2 GPUs). Only hardware-related terms shown; remaining terms consistent with Experiment 1.

Source	% Var	95% CI	Sig.
GPU type (main)	0.12	[0.02, 0.31]	n.s.
`hw×methodology` (agg.)	0.18	[0.03, 0.42]	n.s.
`hw×prompt`	0.05	—	n.s.
`hw×seed`	0.02	—	n.s.
`hw×temp`	0.04	—	n.s.
`hw×ordering`	0.07	—	n.s.
`hw×item`	1.2	[0.7, 1.8]	$p < .05$
`prec×hw`	0.35	—	n.s.
`item_id` (with hw)	59.8	—	—

Table 9: Metric-conditional separability: hardware interaction terms under binary vs. text-exact-match scoring.

Source	Binary (%)	Text-EM (%)
`hw×temp`	0.04	0.83
`hw×prompt`	0.05	—
`hw×seed`	0.02	—
`hw×order`	0.07	—

5.6 Experiment 6: 14B Model Scale Extension

Design. Qwen3-14B and DeepSeek-R1-Distill-14B on MMLU, 200 items, matching the Experiment 1 design (345,600 total records).

Results. Table 14 shows that the variance hierarchy is fully preserved at 14B scale. Item dominance increases slightly from 60.4% (8B mean) to 62.5% (14B mean), a non-significant 2.1 percentage point difference. The increase reflects greater response polarization at higher accuracy (79.2–80.3% vs. 68.5–74.8%). Cross-model model×item between the two 14B models is 22.4%, confirming that model-specific item difficulty persists at larger scale.

DeepSeek-R1-Distill-14B, despite being a reasoning-distilled model, does not alter the MMLU variance structure because MMLU primarily tests knowledge retrieval rather than multi-step reasoning—reasoning distillation's impact would be more pronounced on GSM8K or MATH (future work).

5.7 Experiment 7: Fine-Tuning G-Study

Design. RoBERTa-large fine-tuned on SST-2, MNLI, RTE, and COPA: 6 seeds × 3 learning rates × 4 data shuffles × 3 GPUs × 2 precisions = 432 runs per task, 1,728 total fine-tuning runs with item-level binary predictions.

Regime change. Table 15 reveals a qualitative regime change between inference and fine-tuning variance structures. In inference, item difficulty dominates (60.4%); in fine-tuning, seed×learning-rate interaction becomes the largest source (28.4%), with item difficulty dropping to second place (31.7%). This shift occurs because fine-tuning introduces training-time stochasticity—different random seeds and learning rates can converge to different local minima in the loss landscape, producing models with systematically different decision boundaries on specific items.

Task-specific fine-tuning variance. Table 16 shows that within the fine-tuning regime, item dominance inversely correlates with task difficulty: SST-2 (easiest, accuracy ∼94%) has the lowest item dominance (24.8%) because most items are uniformly easy, while RTE (hardest, accuracy ∼72%) has the highest

Table 10: IRT vs. G-theory item requirements for reliable measurement across four benchmarks.

Benchmark	IRT items	G items	Ratio	G at IRT
MMLU	116	510	4.40×	0.529
HellaSwag	—	—	4.20×	—
ARC-C	—	—	4.35×	—
GSM8K	82	381	4.65×	—
Mean			*4.40×*	

Table 11: Bootstrap 95% CIs on variance components (MMLU, 4-model cross, 1,000 resamples).

Source	Point (%)	95% CI	Excl. 0?
`item_id`	36.8	[33.5, 40.2]	Yes
`model×item`	24.1	[21.3, 27.0]	Yes
Residual	18.8	[16.4, 21.3]	Yes
Model	8.3	[5.1, 12.4]	Yes
Cross-model G	0.065	[0.048, 0.083]	Yes
`infra×method`	0.08	[0.0, 0.42]	**No**

(38.2%). Seed×LR is inversely related, reflecting that training instability matters more when items are less discriminating. GPU×precision remains negligible (0.4%) even in the fine-tuning regime.

5.8 Experiment 8: LayerCast Precision Mitigation

Design. LayerCast (FP16 storage + FP32 compute) added as a fourth precision level on MMLU with Llama-3.1-8B: 1 model × 6 prompts × 6 seeds × 3 temperatures × 4 orderings × 200 items = 86,400 new records.

Results. Table 17 shows that LayerCast reduces precision-related variance by 65%: precision main effect drops from 1.4% to 0.5%, and precision×item from 2.8% to 1.0%. LayerCast accuracy (70.9%) is within 0.3pp of FP32 and 0.2pp of BF16, confirming that FP32 computation recovers most numerical precision. However, the G improvement is modest (+0.023, from 0.618 to 0.641) because precision-related terms constitute only ∼4% of total variance. This confirms that precision is a real but non-dominant variance source—worth mitigating but not the primary lever for improving reproducibility.

5.9 Experiment 9: MATH Benchmark Extension

Design. 4 models × 2 precisions × 6 prompts × 6 seeds × 2 temperatures × 4 orderings × 200 MATH items = 460,800 records.

Results. MATH extends the cross-benchmark gradient to its logical extreme (Table 18). Single-model item dominance drops to 22.1%—the lowest of all five benchmarks—while temperature×item (14.2%) and prompt×item (13.5%) rise to rival item difficulty. The physical mechanism is chain-of-thought amplification: MATH solutions average 200–400 generated tokens (vs. a single letter for MMLU), and each token is a stochastic branching point where sampling noise accumulates. The residual is also elevated (24.6%), reflecting irreducible autoregressive sampling variance in long-form generation.

Cross-model model×item reaches 24.8% on MATH—the highest of all benchmarks—reflecting maximal model-specific reasoning profiles (e.g., Qwen3 excels at algebra while Gemma is stronger at geometry). The extended gradient maintains perfect monotonicity (Spearman $\rho = 1.0$) and expands the spread to 38.3 percentage points. D-study projections require 1,150 items for $G \geq 0.80$ on MATH, 2.4× the MMLU requirement, quantifying the "reasoning variance premium" for proof-level benchmarks.

Table 12: Methodological defense summary (MD1–MD9).

Defense	Test	Result
MD1: Binary data	Null sim. ratio	671×
MD2: Fixed/random	Max Δ item_id	<5% (2.8%)
MD3: Separability scope	Benchmark-qualified	Qualified
MD4: D-study monotonic	Projection check	Monotonic
MD5: G interpretation	G at 14K items	0.943
MD6: Bootstrap indep.	BT/JK CI ratio	0.96–1.04
MD7: IRT vs. G	Item ratio	4.4×
MD8: Design asymmetry	2×2 vs. 3×3	Δ3.6%
MD9: Title alignment	Logprob analysis	Expanded

Table 13: GPU cost estimates for recommended reproducibility budgets.

Benchmark	Items	Prompts	Seeds	GPU-h/model (8B)
MMLU	480	3	4	∼48
HellaSwag	550	3	4	∼62
ARC-C	650	3	5	∼85
GSM8K	900	4	6	∼135

6 Discussion

The benchmark-conditional variance gradient. Our central finding is that variance structure is not a fixed property of "LLM evaluation" but varies systematically with task cognitive complexity. The monotonic gradient (item dominance: 60.4% → 22.1%; prompt×item: 3.2% → 13.5%; $\rho = 1.0$) has a clear mechanistic explanation: knowledge-retrieval tasks have stable, deterministic answers where the model either "knows" or "doesn't know" a fact regardless of prompt phrasing, while multi-step reasoning tasks introduce cumulative sampling variance at each reasoning step. This means any "universal reproducibility recipe" is illusory—a protocol optimized for MMLU (fix prompt, increase items) would be inadequate for GSM8K (must also vary prompts and seeds to capture reasoning instability).

Model×item as a fundamental barrier. The persistence of model×item interaction at 21–25% across all five benchmarks, with cross-model G ranging from 0.035 to 0.065, has a direct implication: leaderboard rankings based on aggregate accuracy are not generalizable to item-level performance. Two models with similar aggregate scores can disagree dramatically on which specific items they answer correctly. D-study projections confirm this is irreducible: even 10 hypothetical models yield $G = 0.179$. The interaction reflects genuine architectural and training-data differences in knowledge coverage and reasoning strategy, not measurement noise.

Conditional separability and its limits. The near-zero infrastructure×methodology interaction (<0.5%) under binary scoring validates the implicit assumption of prior isolated-factor studies: researchers studying prompt sensitivity on one GPU can trust that their findings will not be confounded by hardware choice, at least for binary metrics. However, this assurance is metric-conditional—text-exact-match scoring elevates hardware×temperature to 0.83%—and the hardware×item interaction (1.2%) indicates that a small number of "boundary items" are hardware-sensitive.

The IRT gap. The 4.4× item requirement gap between IRT and G-theory is not a methodological flaw in either framework but reflects fundamentally different reliability targets. IRT asks: "Do we have enough items to stably estimate person ability?" G-theory asks: "Do we have enough items to generalize across *all* evaluation conditions (prompts, seeds, orderings)?" At IRT's recommended 116 items, G-theory G is only 0.529 because prompt×item, seed×item, and residual variance—invisible to IRT—erode reliability. This gap

Table 14: 8B vs. 14B variance decomposition comparison (binary, MMLU). Δ is 14B mean − 8B mean.

Source	8B mean	Qwen3-14B	DS-R1-14B	14B mean
item_id	60.4	63.8	61.2	62.5
prompt×item	3.2	3.9	3.7	3.8
seed×item	3.1	4.2	4.0	4.1
Residual	21.7	19.2	19.8	19.5
G	0.618	0.651	0.633	0.642

Table 15: Inference vs. fine-tuning variance decomposition (binary). Inference: 4-model MMLU mean. Fine-tuning: RoBERTa-large mean across 4 NLU tasks.

Inference		Fine-tuning	
Source	%	Source	%
item_id	**60.4**	seed×lr	**28.4**
Residual	21.7	item_id	31.7
pr×i	3.2	shuffle×i	12.3
se×i	3.1	Residual	12.1
te×i	2.1	LR (main)	8.9
pc×i	1.8	Seed (main)	6.2
Precision	1.4	gpu×prec	0.4

grows with reasoning complexity (4.2× for HellaSwag to 4.65× for GSM8K) as non-item variance sources strengthen.

Inference vs. fine-tuning regimes. The qualitative regime change from item-dominated (inference) to seed×LR-dominated (fine-tuning) variance implies that reproducibility prescriptions must be regime-specific. Inference reproducibility is primarily an evaluation design problem (more items, multiple prompts); fine-tuning reproducibility is a training stability problem (seed/LR sensitivity, data ordering effects). Our finding complements Dodge et al. [2020] and Sellam et al. [2022] by providing the first G-theory decomposition of fine-tuning variance with explicit comparison to inference-time structure.

Limitations. Our analysis has several limitations. (1) Binary G-theory assumes continuous responses; we mitigate this with null simulations (observed/null ratio = 671×) and parallel logprob analyses. (2) Temperature and ordering facets have limited design points (3 and 4 respectively); fixed-vs-random sensitivity analysis shows <3% impact. (3) The fine-tuning experiment uses encoder-only RoBERTa; decoder fine-tuning (LoRA on LLMs) requires substantially more compute and is deferred. (4) Only open-weight models are studied; API models have opaque infrastructure that prevents controlled experimentation. (5) 14B models require tensor parallelism, introducing cross-GPU communication nondeterminism as an additional unmeasured variance source.

7 Conclusion

We presented the first multi-benchmark G-theory study unifying infrastructure, methodology, and measurement variance in LLM evaluation. Three findings emerge. First, model×item interaction (21–25%) is a fundamental, benchmark-modulated barrier to cross-model generalizability that cannot be reduced by adding more models ($G = 0.179$ even with 10 models). Second, infrastructure×methodology separability is confirmed under binary scoring (<0.5%) but is metric-conditional. Third, the variance structure itself follows a monotonic gradient from knowledge to reasoning benchmarks ($\rho = 1.0$), with reasoning benchmarks requiring 1.9× the item budget of knowledge benchmarks for equivalent reliability. G-theory requires 4.4× more items than IRT because it accounts for prompt, seed, and ordering dimensions that IRT ignores. We

Table 16: Fine-tuning variance decomposition by NLU task (RoBERTa-large, binary).

Source	SST-2	MNLI	RTE	COPA
`item_id`	24.8	33.5	38.2	30.4
`seed`×`lr`	32.1	25.8	26.4	29.2
`shuffle`×`i`	—	—	—	—
Residual	—	—	—	—

Table 17: LayerCast impact on precision-related variance components and generalizability (MMLU, Llama, binary).

Component	Before	After	Δ
Precision (main)	1.4%	0.5%	−64%
`prec`×`item`	2.8%	1.0%	−64%
Other sources	<0.3pp change		—
G	0.618	0.641	+0.023

release an open-source toolkit for computing benchmark-specific reproducibility budgets, translating these findings into actionable evaluation design guidance.

References

Bouthillier, X., Delaunay, P., Bronzi, M., Trofimov, A., Nichyporuk, B., Szeto, J., ... & Pal, C. (2021). Accounting for variance in machine learning benchmarks. *Proceedings of Machine Learning and Systems*, 3.

Brennan, R. L. (2001). *Generalizability Theory*. Springer.

Brittlebench Authors. (2026). Brittlebench: Quantifying prompt brittleness in LLM evaluation. *Preprint*.

Cronbach, L. J., Gleser, G. C., Nanda, H., & Rajaratnam, N. (1972). *The Dependability of Behavioral Measurements*. Wiley.

Dehghani, M., Tay, Y., Gritsenko, A. A., Zhao, Z., Houlsby, N., Diaz, F., ... & Metzler, D. (2021). The benchmark lottery. *arXiv preprint arXiv:2107.07002*.

Dodge, J., Ilharco, G., Schwartz, R., Farhadi, A., Hajishirzi, H., & Smith, N. (2020). Fine-tuning pretrained language models: Weight initializations, data orders, and early stopping. *arXiv preprint arXiv:2002.06305*.

Henderson, P., Islam, R., Bachman, P., Pineau, J., Precup, D., & Meger, D. (2018). Deep reinforcement learning that matters. *Proceedings of AAAI*.

Hua, W., et al. (2025). Disentangling evaluation artifacts from genuine prompt sensitivity. *Proceedings of EMNLP*.

Kunievsky, A. (2025). Decomposing LLM evaluation uncertainty: Intent, articulation, and model factors. *Preprint*.

Lalor, J. P., Wu, H., & Yu, H. (2019). Learning latent parameters without human response patterns: Item response theory with artificial crowds. *Proceedings of EMNLP*.

Lu, Q., et al. (2022). Are emergent abilities in large language models just in-context learning? A study of option ordering effects. *Preprint*.

Mesting, F. (2026). Generalizability theory for LLM evaluation: A five-facet analysis. *Preprint*.

Table 18: Extended 5-benchmark variance gradient (single-model, binary). Includes D-study item requirements for $G \geq 0.80$.

Source	MMLU	Hella.	ARC-C	GSM8K	MATH
`item_id`	60.4	50.4	42.1	28.3	**22.1**
`pr`×`i`	3.2	5.4	7.3	11.8	13.5
`te`×`i`	2.1	5.2	6.5	9.4	14.2
`se`×`i`	3.1	4.9	5.8	8.2	9.8
Residual	21.7	21.3	20.5	18.2	24.6
G	0.618	0.548	0.495	0.385	0.328
$\text{Items}_{G\geq.80}$	480	550	650	900	1150
Cross-model G	0.065	0.058	0.052	0.041	0.035
`m`×`i` (%)	24.1	21.3	22.7	23.5	24.8

Messina, P. & Scotta, A. (2026). Background temperature: Characterizing non-determinism in LLM inference. *Transactions on Machine Learning Research*.

Perlitz, Y., et al. (2023). Efficient benchmarking (of language models). *arXiv preprint*.

Pezeshkpour, P. & Hruschka, E. (2024). Large language models sensitivity to the order of options in multiple-choice questions. *Findings of ACL*.

Polo, F. M., et al. (2024). tinyBenchmarks: Evaluating LLMs with fewer examples. *Proceedings of ICML*.

Rodriguez, P., Barrow, J., Hoyle, A. M., Lalor, J. P., Jia, R., & Boyd-Graber, J. (2021). Evaluation examples are not equally informative: How should that change NLP leaderboards? *Proceedings of ACL*.

Sellam, T., Yadlowsky, S., Wei, J., Saphra, N., D'Amour, A., Linzen, T., ... & Eisenstein, J. (2022). The MultiBERTs: BERT reproductions for robustness analysis. *Proceedings of ICLR*.

Shavelson, R. J. & Webb, N. M. (1991). *Generalizability Theory: A Primer*. Sage.

ThinkingMachines Authors. (2025). Batch-invariance failures in LLM inference: Sources and mitigations. *Preprint*.

Vania, C., et al. (2021). Comparing test sets with item response theory. *Proceedings of ACL*.

Yuan, Z., et al. (2025). Numerical precision and LLM evaluation: Up to 9% accuracy variation from floating-point format. *Preprint*.

Biderman, S., et al. (2024). Lessons from the trenches on reproducible evaluation of language models. *Preprint*.